

CURRICULUM VITÆ: FRANÇOIS MASSOL

Unit Evo-Eco-Paléo, SPICI group
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Born September 17th, 1979
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[RESEARCHGATE](#); [MENDELEY](#); [ACADEMIA](#); [ORCID](#); [RESEARCHER ID](#); [GOOGLE SCHOLAR](#); [SCOPUS](#)

Recent professional positions

- 2012 – **Permanent researcher** CNRS, [CEFE](#) Montpellier (2012 – 2014), [EEP](#) Lille (2013 –)
Hired through a national competition, with ~6 posts for ~200 junior applicants in ecology/evolution.
- 2009 – 2010 **Marie Curie post-doc** at the University of Texas, Austin, USA (supervisor: M. Leibold)
- 2008 – 2012 **Permanent researcher** IRSTEA, Aix-en-Provence (2008 – 2009 and 2010 – 2012)
- 2002 – 2008 **Engineer / Civil servant** ENGREF, Paris

Education

- 2015 **Habilitation (HDR)** in Biodiversity Science at the University of Lille
HDR is the diploma allowing researchers to be PhD student main supervisors; ~ tenure or privatdozent.
- 2004 – 2008 **PhD** in Population Biology & Ecology at the University of Montpellier
Supervised by D. Gerdeaux and P. Jarne; paid as civil servant for the Ministry of Agriculture.
- 2003 – 2004 **Master** in Evolutionary Biology & Ecology at the University of Montpellier
- 1999 – 2003 **Interdisciplinary graduate studies** at Ecole Polytechnique and ENGREF, 2 MSc degrees

Advising & supervision

- 2013 – Member of 8 PhD advisory committees (Montpellier, Toulouse, La Réunion, Marseille, Dijon)
- 2008 – Adviser or co-adviser of 7 post-docs, 3 PhD students, 14 MSc students
1 PhD student defended his thesis (F. Laroche) and is now a permanent researcher in France.
5 former post-docs now have permanent or tenure-track positions in Europe, Argentina or the US (K. Allhoff, J. Astegiano, J. Pantel, M. Pantasso, M. Thomas).

Teaching

- 2014 **Invited graduate course** – evolutionary dynamics, SAIIR, São Paulo, Brazil
- 2013 **Invited graduate course** – metacommunity theory, ETH/Uni. Zürich, Ascona, Switzerland
- 2013 – **Regular lectures at MSc level** (10 – 16h per year) – ecological modelling, evolutionary biology, methods for the study of evolution, the publication process, etc., Univ. Lille

Funding

As principal investigator (PI)

- 2009 – 2011 **DEFTER** Post-doctoral fellowship (EU Marie Curie programme) on “dispersal effects on food web functioning: theory and experimental research based on freshwater plankton systems” (in M. Leibold’s laboratory in Austin, TX)
- 2010 – 2012 **FoodWebs** Working group at the National Institute for Mathematical and Biological Synthesis (NIMBioS) on “food web dynamics and stoichiometric constraints in metaecosystems” (PIs: C. Klausmeier, M. Leibold, F. Massol & R. Sterner)
- 2013 – 2018 **MIRES** Working group funded by the Réseau National Systèmes Complexes (RNSC) and the Institut National de la Recherche Agronomique (INRA) applied mathematics and informatics department on “interdisciplinary methods for the study of seed exchange networks” (PIs: P. Barbillon, S. Caillon, F. Massol, M. Thomas & N. Verzealen)
- 2013 – 2014 **DyBRES** Working group funded by the RNSC on “biodiversity dynamics in spatial ecological networks” (PIs: S. Manel, F. Massol & F. Munoz)
- 2014 – 2017 **COREIDS** Working group at the Centre de Synthèse et d’Analyse de la Biodiversité (CESAB) on “Predicting community resilience to invasions from diversity and network structure” (PIs: P. David & F. Massol)
- 2014 **RESEDA** Bilateral cooperation (funded by the University of Lille 1) with the University of Exeter on “selection regimes, dispersal and local adaptation” (PIs: F. Débarre & F. Massol)

2014 – 2018	ARSENIC	Project funded by the Agence Nationale de la Recherche (ANR) on “adaptation and resilience of spatial ecological networks to human-induced changes” (PIs: N. Loeuille & F. Massol)
2016 – 2017	FEEDING	Interregional project Ghent-Louvain-Lille on “functional and evolutionary ecology of dispersal: integrative approaches under global change” (PIs: N. Schtickzelle, D. Bonte & F. Massol)
2017 – 2018	E²-SPACE	Interregional project Ghent-Louvain-Lille on “eco-evolutionary dynamics in space” (PIs: N. Schtickzelle, D. Bonte & F. Massol)

Others

2011 – 2014	NETSEED	Working group at the CESAB on “seed exchange networks” (PI: D. McKey)
2012 – 2017	AFFAIRS	Project funded by the ANR BIOADAPT programme on “adaptation of fragmented freshwater assemblages in rapidly-changing environments” (PI: P. David)
2015 – 2018	AREOLAIRE	Project funded by the Région Nord Pas-de-Calais / FRB programme on “adaptation, regression and expansion at species range margins” (PI: A. Duputié)
2015 – 2016	MADRES	Project funded by the CNRS interdisciplinary mission MoMIS programme on “modelling and analysis of the dynamics of seed exchange networks” (PI: S. Martin)
2018 – 2019	sEcoEvo	Working group at the sDiv synthesis centre (Berlin, Germany) on “Biodiversity Dynamics – The Nexus Between Space & Time” (PIs: R. G. Gillespie & M. Hickerson)
2018 – 2021	SUBANTECO	Project funded by the Institut polaire français Paul-Emile Victor (IPEV) on “Subantarctic biodiversity, effects of climate change and biological invasions on terrestrial biota” (PI: D. Renault)
2017 – 2021	NGB	Project funded by the ANR on “next-generation biomonitoring of change in ecosystem structure and function” (PI: D. Bohan)

Awards

2016 – 2020	Supervision and Research Bonus (PEDR), granted by the CNRS
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Commissions of trust

2017 –	Founder, manager and recommender for <i>Peer Community in Ecology</i>
2017 –	Participation in researcher and lecturer hiring committees (Univ. Paris Diderot, IRSTEA)
2016 – 2017	Guest volume editor for the 56 th and 57 th volumes of <i>Advances in Ecological Research</i>
2015 –	Member of the administrative board of the French Society of Ecology and Evolution (SFE ²)
2015 –	Referee at 1 HDR defence; examiner at 2 HDR defences (Montpellier, Toulouse, Lille)
2014 –	Referee at 1 PhD defence; examiner at 5 PhD defences (Paris, Dijon, Montpellier, Ecole Polytechnique)
2014 –	Subject editor for the journal <i>Oikos</i>
2011 –	Reviewer of proposals for funding agencies (Austria, Belgium, France, Israel, Russia, Switzerland)
2008 –	Reviewer for > 20 journals including <i>PNAS</i> , <i>Proc. R Soc B</i> , <i>Ecol. Lett.</i> , <i>Nat. Commun.</i> , <i>Nat. Ecol. Evol.</i>

Since 2007, I have authored **55 peer-reviewed articles** (**14 as first author, 9 as last author**), in major multidisciplinary (1 *Nature*, 2 *Nature Communications*, 3 *Scientific Reports*, 1 *Science Advances*) and general biology, ecology and evolutionary biology journals (2 *Trends in Ecology and Evolution*, 4 *Ecology Letters*, 1 *Biological Reviews*, 1 *Ecological Monographs*, 1 *Methods in Ecology and Evolution*, 2 *Proceedings of the Royal Society B*, 4 *Evolution*, 1 *Journal of Ecology*...). I have also contributed to **1 book chapter** and edited **2 books** (volumes 56 and 57 of the *Advances in Ecological Research* series). I gave **2 invited plenary lectures**, **7 invited oral communications**, **5 oral contributions**, 21 oral presentations in workshops and 16 seminars in various universities.

Citations: 1682 (Google Scholar) / 1104 (ISI); **1 highly cited paper** (on ISI web of science¹)

H-index: 22 (Google Scholar) / 18 (ISI)

All articles can be found here: <https://sites.google.com/a/polytechnique.org/francoismassol/publications>

In the following, students are underlined. * indicates co-first authorship. Impact factors (IF) were obtained from the 2016 ISI Journal Citation Reports, numbers of citations were obtained from Web of Science on 25/05/2018.

¹ received enough citations to place them in the top 1% of their academic field based on a highly cited threshold for the field and publication year.

Representative peer-reviewed publications (non-exhaustive list)

- (1) Gravel, D.*, **Massol, F.***, & Leibold, M. (2016) Stability and complexity in model meta-ecosystems. *Nature Communications*, **7**, 12457. **11 citations; journal IF: 12.124**
- (2) **Massol, F.** & Débarre, F. (2015) Evolution of dispersal in temporally variable environments: the importance of life cycles. *Evolution*, **69**, 1925-1937. **5 citations; journal IF: 4.201**
- (3) **Massol, F.**, Dubart, M., Calcagno, V., Cazelles, K., Jacquet, C., Kéfi, S., & Gravel, D. (2017) Island biogeography of food webs. *Advances in Ecological Research*, **56**, 183-262. **7 citations; journal IF: 5.056**
- (4) **Massol, F.**, Gravel, D., Mouquet, N., Cadotte, M. W., Fukami, T., & Leibold, M. A. (2011) Linking ecosystem and community dynamics through spatial ecology. *Ecology Letters*, **14**, 313-323. **99 citations; journal IF: 9.449**
- (5) Gravel, D., **Massol, F.**, Canard, E., Mouillot, D., & Mouquet, N. (2011) Trophic theory of island biogeography. *Ecology Letters*, **14**, 1010-1016. **77 citations; journal IF: 9.449**
- (6) **Massol, F.**, Duputié, A., David, P., & Jarne, P. (2011) Asymmetric patch size distribution leads to disruptive selection on dispersal. *Evolution*, **65**, 490-500. **32 citations; journal IF: 4.201**
- (7) Laroche, F., Jarne, P., Lamy, T., David, P., & **Massol, F.** (2015) A neutral theory for interpreting correlations between species and genetic diversity in communities. *American Naturalist*, **185**, 59-69. **14 citations; journal IF: 4.167**
- (8) Tasiemski, A., **Massol, F.**, Cuvillier-Hot, V., Boidin-Wichlacz, C., Roger, E., Rodet, F., Fournier, I., Thomas, F., & Salzet, M. (2015) Reciprocal immune benefits based on the complementary production of antibiotics by the leech *Hirudo verbana* and its gut symbiont *Aeromonas veronii*. *Scientific Reports*, **5**, 17498. **7 citations; journal IF: 4.259**
- (9) Cheptou, P.-O. & **Massol, F.** (2009) Pollination fluctuations drive evolutionary syndromes linking dispersal and mating system. *American Naturalist*, **174**, 46-55. **54 citations; journal IF: 4.167**
- (10) Duputié, A. & **Massol, F.** (2013) An empiricist's guide to theoretical predictions on the evolution of dispersal. *Interface Focus*, **3**. **31 citations; journal IF: 2.693**
- (11) Calcagno, V.*, **Massol, F.***, Mouquet, N., Jarne, P., & David, P. (2011) Constraints on food chain length arising from regional metacommunity dynamics. *Proceedings of the Royal Society Biological Sciences Series B*, **278**, 3042-3049. **25 citations; IF: 4.940**
- (12) Schleuter, D., Daufresne, M., **Massol, F.**, & Argillier, C. (2010) A user's guide to functional diversity indices. *Ecological Monographs*, **80**, 469-484. **236 citations; IF: 8.759**

Other representative publications (non-exhaustive lists)

Open review publications

Lavoir, A.-V., Staudt, M., Schnitzler, J. P., Landais, D., **Massol, F.**, Rocheteau, A., Rodriguez-Cortina, R., Zimmer, I., & Rambal, S. (2009) Drought reduced monoterpene emissions from *Quercus ilex* trees: results from a throughfall displacement experiment within a forest ecosystem. *Biogeosciences Discussions*, **6**, 863-893.

Edited books

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: A Synthesis of Concepts*. Vol. 56, *Advances in Ecological Research* series.

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: Empirical Evidence and Case Studies*. Vol. 57, *Advances in Ecological Research* series.

Book chapters

Massol, F., Jabot, F., Manel, S., & Munoz, F. (2015) Etude des réseaux en écologie. In: *Les Réseaux dans les Sociétés Animales* (ed.: Sueur, C.) Editions Matériologiques, 49-98.

Science popularisation & expertise

Massol, F. & Chave, J. (2017) Ecologie fondamentale : les grands défis. In: *Prospectives de l'Institut Ecologie & Environnement du CNRS, Compte-rendu des journées des 22, 23 et 24 Février 2017, Bordeaux*, 7-13.

Invited presentations

Plenary talks at international conferences

- 2016 Networks of interactions in ecology: Everything you always wanted to know about ecological networks (but were afraid to ask). *Société Française d'Ecologie internationale conference*, Marseille, France
- 2014 Individual dispersal, fluxes of nutrients and the stability of ecosystems. *SysBio conference*, Lyon, France

Invited talks at international symposia

- 2017 The evolution of conditional dispersal under different life cycles. *CIMI conference: Ecology and evolutionary biology, deterministic and stochastic models*, Toulouse, France
- 2016 Life-history traits and the properties of metaecosystems: a theoretical perspective. *Per Brink thematic symposium, Nordic Oikos conference*, Turku, Finland
- 2016 Diversification of dispersal rates in variable environments. *Maths-Bio conference day*, Paris, France
- 2015 Joint evolution of dispersal and the allocation of dispersal cost in heterogeneous landscapes. *Mathematical Models in Ecology and Evolution*, Paris, France
- 2014 Community dynamics and the evolution of dispersal. *ECMTB meeting*, Göteborg, Sweden
- 2014 Rescuing metacommunity ecology using random matrix theory. *Journées Modélisation Aléatoire et Statistique*, Toulouse, France
- 2010 Recasting spatial food web ecology as an ecosystem science. *ESA meeting*, Pittsburgh, USA

Other representative presentations (non-exhaustive lists)

Contributed oral communications

- 2017 Eco-evolutionary dynamics of colonization rate, patch occupancy and food chain length in model metacommunities. *ESEB meeting*, Groningen, the Netherlands, with V. Calcagno
- 2017 Effects of cue reliability, environmental uncertainty and the life cycle on the evolution of emigration strategies. *British Ecological Society meeting*, Ghent, Belgium, with F. Débarre
- 2011 Evolution in temporally variable environments: the importance of life cycles. *Mathematical & Theoretical Ecology conference*, Colchester, UK, with F. Débarre

Talks at workshops

- 2017 Modelling the dynamics of spatially structured ecosystems. *Ecosystem Theory conference*, Montpellier
- 2017 Dispersal and the stability of ecosystems. *Hausdorff School: Large random graphs: geometry and applications*, Bonn, Germany
- 2017 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *FEEDING workshop*, Ghent, Belgium
- 2012 Metapopulation-based approaches to spatial food webs. *Networks and interactions in agriculture workshop*, Dijon

Seminars

- 2018 The ecology and evolution of food-chain length in metacommunities. *Sherbrooke, Canada*
- 2017 Evolution of dispersal in spatially and temporally variable environments. *Louvain-la-Neuve, Belgium*
- 2016 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *Helsinki, Finland*
- 2015 Stability, dispersal and ecological networks. *Ecole Polytechnique, Palaiseau, France*
- 2014 A model for the evolution of dispersal in heterogeneous environments. *Ghent, Belgium*
- 2013 Adaptive dynamics in a heterogeneous and uncertain world. *Montpellier, France*
- 2012 The evolution of dispersal: old questions, new answers. *Paris, France*